EXPERIMENT-4

4. Scenario: You are working on a project that involves analyzing the sales performance of a company over the past four quarters. The quarterly sales data is stored in a NumPy array named sales_data, where each element represents the sales amount for a specific quarter. Your task is to calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter.

Question: Using NumPy arrays and arithmetic operations calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter?

Aim

To calculate the total sales for four quarters and determine the percentage increase in sales from the first quarter to the fourth quarter using NumPy.

Algorithm

1. Import the NumPy library.
2. Create a NumPy array containing sales for Q1, Q2, Q3, and Q4.
3. Calculate the total yearly sales using np.sum().
4. Calculate the percentage increase from Q1 to Q4.
5. Display the total sales and percentage increase.

Code

```
import numpy as np

sales_data = np.array([100000, 120000, 130000, 150000])

total_sales = np.sum(sales_data)

percentage_increase = ((sales_data[3] - sales_data[0])

                       / sales_data[0]) * 100

print("Total sales for the year:", total_sales)
```

```
print("Percentage increase from Q1 to Q4:",
      percentage_increase, "%")
```

Input

Q1 = 100000

Q2 = 120000

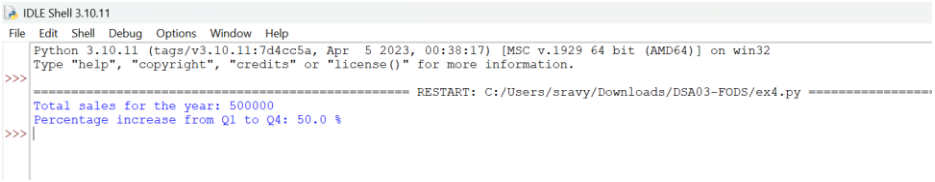
Q3 = 130000

Q4 = 150000

Output

Total sales for the year: 500000

Percentage increase from Q1 to Q4: 50.0 %



Result

Thus, the total yearly sales were calculated as **500000**, and the percentage increase from Q1 to Q4 was calculated as **50%**.